

Topology

Theatre nodes (×12)

Each theatre is its own subnet, NAT'd by its own GL-iNet **Slate AX (GL-AXT1800)** router (previously the A-1300 / Slate Plus; replaced 2026-10-06 because Tailscale's userspace WireGuard on the A-1300's 32-bit CPU was estimated at only ~30-70 Mbps — see docs/bandwidth-analysis.md). The router runs Tailscale as a subnet router (LAN + WAN side routing enabled), advertising the theatre's /24 .

Physical wiring: the Slate AX has 1 WAN + 2 LAN gigabit ports (the same layout the A-1300 had, so the wiring is unchanged). Both LAN ports are used:

- **LAN 1 → ATEM Mini Extreme ISO directly.** Dedicated port, so the router's other LAN traffic (rclone syncs, control laptop, etc.) never shares a switch segment with it. The ATEM carries the heaviest sustained traffic of any theatre device — the near-real-time ingest pull (see docs/atem-iso-ingest.md) — camera ISO + program by default — is a continuous ~47-83 Mbps flow, well above BirdDog Play's — so it gets the dedicated port.
- **LAN 2 → 8-port unmanaged Netgear switch,** which fans out to everything else: BirdDog Play, PowerPoint Main/Backup, VT Main/Backup, Control laptop (6 devices + 1 uplink port, 1 spare switch port).

All devices stay on the same 192.168.X.0/24 — this is a physical port split for traffic separation, not a VLAN/subnet split.

Role	Address	Physical connection
GL-iNet Slate AX router (LAN gateway)	192.168.X.1	—
ATEM Mini Extreme ISO	192.168.X.2	Slate AX LAN 1 (direct)
BirdDog Play	192.168.X.20	Netgear switch → Slate AX LAN 2
PowerPoint Main	192.168.X.5	Netgear switch → Slate AX LAN 2
PowerPoint Backup	192.168.X.6	Netgear switch → Slate AX LAN 2
VT Main	192.168.X.7	Netgear switch → Slate AX LAN 2
VT Backup	192.168.X.8	Netgear switch → Slate AX LAN 2
Control laptop	192.168.X.10	Netgear switch → Slate AX LAN 2

X per the subnet map in the top-level README: Theatre 1 → 2 , Theatre 2 → 3 , ... Theatre 12 → 13 .

The ATEM's built-in FTP server (its ISO recording drive) is pulled continuously by the mothership over Tailscale subnet routing — see docs/atem-iso-ingest.md .

Alternative to the GL-iNet router, documented but not recommended: plain switch + Tailscale on generic Linux — see docs/topology-alternative-tailscale-switches.md .

Uplink VLAN grouping

Theatres are grouped 3-per-physical-VLAN purely to organize uplink cabling back to the mothership — **this grouping does not itself provide cross-theatre isolation**. Each GL-iNet Slate AX already NATs its own theatre LAN, and once every router joins the same tailnet, routes are reachable tailnet-wide unless Tailscale ACLs restrict them (see `config/tailscale-acl.json`).

These VLANs are venue-supplied, not our infrastructure — each capped at 1 Gbps, each port expensive. The 4-VLAN/3-theatres-each split below is comfortable at every live-ingest scope; the bandwidth math allows consolidating to 2 VLANs (6 theatres each) for normal SRT-primary operation at the default camera + program scope (~50% typical / ~65% high upstream, higher in the group carrying both VMix nodes) or camera + 1 laptop at typical rates (~65%), but only ~10% headroom in a mass NDI fallback — see `docs/bandwidth-analysis.md` for the full model and recommendation.

VLAN	Theatres	Subnets
1	1, 2, 3	.2.x, .3.x, .4.x
2	4, 5, 6	.5.x, .6.x, .7.x
3	7, 8, 9	.8.x, .9.x, .10.x
4	10, 11, 12	.11.x, .12.x, .13.x

Mothership — 192.168.1.x

Ubiquiti **Cloud Gateway**; all 4 VLANs terminate here via separate links. Also hosts the 4× "ready room" computers (physical, untouched) and the hardwired WAN.

Consolidated services server (Unraid). Everything else that used to be described as separate mothership boxes — Nextcloud, Restreamer, BirdDog Central, the NDI Discovery Server, a self-hosted DERP server, the ATEM ISO ingest pipeline, the VMix record ingest pipeline, a self-hosted UniFi Controller, self-hosted GLKVM-Cloud, ATEM Overseer + ATEM Fleet Admin (fleet monitoring and provisioning for the 12 theatre ATEMs), and now Flock (BirdDog Play fleet manager) — runs on a single physical server instead, virtualized with Unraid. (An earlier revision also carried a mothership-local VMix instance VM; it was removed from the design — the theatre program feeds originate from the VMix *node* PCs, and no mothership role for a local VMix instance was ever established, see `docs/open-questions.md` #10.) The server already exists (no procurement needed) — known so far: **2× gigabit NICs (bonded — see below), a "decent" discrete GPU** (on hand, though nothing in the design requires it any more — see the GPU section of `docs/server-specification.md`). Full spec (CPU/motherboard, RAM) is TBD — see `docs/open-questions.md` and the calculated target minimum spec (storage pool layout, RAM, CPU, GPU tier, all worked from this box's actual workload) in `docs/server-specification.md` .

Workload	Runs as	Notes
BirdDog Central	Windows VM	Windows-only (tech specs); the design's only VM — retained for the NDI backup path's sender side (<code>docs/streaming-flow.md</code>), not for fleet management (that's Flock's job)
Nextcloud	Docker container	official image + MariaDB + Redis (see <code>config/docker-compose.yml</code>)
Restreamer	Docker container	official <code>datarhei/restreamer</code> image
NDI Discovery Server	Docker container	see <code>docs/streaming-flow.md</code>

Workload	Runs as	Notes
DERP server	Docker container	self-hosted relay fallback — see <code>docs/tailscale.md</code>
ATEM ISO Ingest	Docker container	pulls all 12 theatres' ATEM ISO recordings via FTP, dual-writing to Nextcloud and the edit-suite NAS in the same pass — see <code>docs/atem-iso-ingest.md</code> and <code>docs/live-editing.md</code>
VMix Record Ingest	Docker container	pulls all 4 VMix PCs' recordings via SMB, same dual-write pattern — see <code>docs/vmix-record-ingest.md</code>
UniFi Controller	Docker container	self-hosted UniFi Network Application — local management/backup console for the Cloud Gateway, independent of cloud.ui.com — see <code>config/unifi/README.md</code>
GLKVM-Cloud (<code>rttys</code> + <code>coturn</code>)	2 Docker containers	self-hosted remote administration (web UI + SSH terminal) for the 14 GL-iNet routers — see <code>docs/glkvcloud.md</code>
ATEM Overseer	Docker container	fleet monitoring/tally dashboard for all 12 theatres' ATEMs — receives each theatre's monitoring stream, see <code>docs/bandwidth-analysis.md</code>
ATEM Fleet Admin	Docker container	bulk ATEM provisioning — model-aware config forms, live network apply via <code>atem-connection</code> over the same Tailscale subnet routing the ISO ingest pipeline uses, or XML+media folder export
Flock	Docker container	BirdDog Play fleet manager — LAN discovery, tag-based grouping, BirdUI-parity settings, batch edits; receives each theatre's BirdDog Play preview stream, see <code>docs/bandwidth-analysis.md</code>
Tailscale subnet router	Docker container, host networking	advertises <code>192.168.1.0/24</code> ; replaces the earlier "Nextcloud host" plan now that Nextcloud itself is a container on this same box

Bonded NICs. Decided: bond the 2 gigabit NICs (802.3ad/LACP) rather than split traffic across them — SRT bandwidth isn't constant, and LACP's hash-based distribution spreads the many *simultaneous* flows this box handles (12 theatres' SRT fan-out, each to a different destination, the 12 incoming Overseer monitoring streams and Flock SRT previews, plus rclone/Nextcloud/BirdDog Central/NDI Discovery Server/DERP traffic) across both links by destination-IP/port hash — giving real aggregate headroom for bursty traffic, not just failover. Note this does **not** speed up any single flow; the benefit comes from having many distinct flows to hash across. On the wire, though, everything bound for one theatre leaves the Tailscale container inside a single WireGuard UDP flow to that theatre's Slate AX, so the bond actually hashes ~14 tunnel flows (one per router), not dozens of application flows — still enough to spread, but coarse: one link can end up carrying noticeably more than half.

Assuming the Cloud Gateway supports LAG on the ports the Unraid server lands on (only UDM Pro/SE/Pro Max, UXG Enterprise, and EFG support port aggregation — not base Cloud Gateway models). If it turns out this one doesn't: drop an intermediate LACP-capable switch in between and bond through that instead — the bond doesn't care which device terminates it, only that something in the path speaks LACP, so this is a cheap fallback rather than a blocker.

One thing worth doing regardless of which device terminates the bond: **set Unraid's bond to match Ubiquiti's LACP settings.** Ubiquiti gear hardcodes LACP rate `fast` and hash policy `layer3+4` ; Unraid's bonding defaults are `slow` rate and `layer2` hash. The bond still forms with a mismatched hash policy (each end hashes its own transmit traffic independently), but `layer2` hashing on the Unraid side puts everything bound for the one gateway MAC on a single link, which throws away the outbound headroom the bond exists for.

Why Unraid over TrueNAS SCALE: both support Docker and Windows VMs, and Unraid has the more polished one-click container experience (Community Applications) — useful since this may be maintained on-site by AV staff rather than a Linux admin. Trade-off: Unraid requires a paid license (2026 tiers: Starter \$49 for 6 storage devices, Unleashed \$109, Lifetime \$249 — see `docs/open-questions.md`

for which tier this box's pools need); TrueNAS SCALE is free. (The original tiebreaker — Unraid's more mature GPU-passthrough workflow — no longer applies now that the VMix VM is gone: nothing on this box needs GPU passthrough. BirdDog Central is NDI routing/control, not video encode, and runs as a plain VM. The choice stands on the container/UX grounds alone.)

Tailscale note: Cloud Gateway (UniFi-OS-based) generally can't run Tailscale natively without an unofficial container hack, so the mothership subnet-router role runs as a container on the Unraid box instead of the Cloud Gateway itself. See `docs/open-questions.md`.

VMix nodes (×2)

Each has its own **GL-iNet Slate AX** router (same model as the theatre routers) and its own Tailscale connection — a separate node on the tailnet, not part of the theatre it physically sits next to, just uplinked near it:

Node	Uplinks near	Subnet	Contents
VMix Node 1	Theatre 1	192.168.20.x	4× BirdDog P400 cameras, 2× VMix PCs
VMix Node 2	Theatre 4	192.168.21.x	4× BirdDog P400 cameras, 2× VMix PCs

That makes **14 GL-iNet Slate AX routers total** across the network — 12 theatre routers plus these 2. Both LAN ports are bridged together on the VMix node routers (unlike the theatre routers, there's no single traffic-sensitive device here that needs its own dedicated port the way the ATEM does).

Each VMix PC's own recording is pulled to the mothership the same way the ATEMs' recordings are — near-real-time, over this node's own dedicated uplink (not the adjacent theatre's) — see `docs/vmix-record-ingest.md`.

Edit suite — 192.168.22.x

Post-production subsystem at the mothership: 2× MacBook Pro editing workstations, a Mac mini running the DaVinci Resolve Project Server and Remote Render, and a dedicated fast-storage NAS, on their own 10GbE LAN — **not** on the Tailscale mesh (same room as the mothership, no distance for Tailscale to bridge) and not sharing the theatre-facing network's bandwidth budget. Routed to the mothership LAN only so the ingest containers can dual-write footage to the edit-suite NAS in the same pull that feeds Nextcloud — there is no separate recording-pool-to-NAS sync step. Full design, including the storage and project-server decisions and the DIT physical-media ingest workflow, in `docs/live-editing.md`.

See `diagrams/topology.svg` for the full visual layout — every theatre, every device, every real IP — and `docs/ip-address-map.md` for the same inventory as a text table. Per-device configs: all 14 routers in `config/gl-inet/`, the Cloud Gateway in `config/unifi/`, and Tailscale in `config/tailscale-up-all-devices.sh`.